Using AI to Build Interactive Teaching Materials

From Conflict to Collaboration

Dr. Irving Gómez Méndez

Teaching in the AI Era: Build Your AI Teaching Toolkit

6 November 2026

What you get today

You build one working simulation

An HTML file for a topic you teach. It opens in any browser.

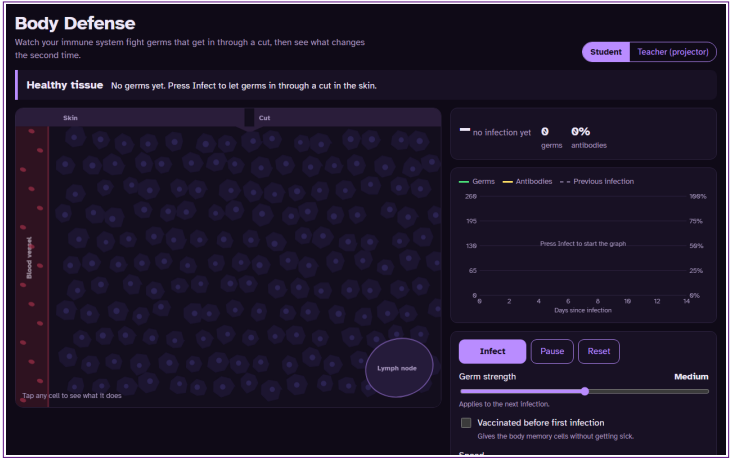
You take home a kit

Prompts, examples and a check list, so you can build more.

The kind of thing we build today:

gizmos.explorelearning.com

Body Defense: an immune system simulation



Built by Dr. Irving Gómez Méndez

Where are you today?

Which sentence is closest to you?

A

I want to stop my students from using AI.

Where are you today?

Which sentence is closest to you?

A

I want to stop my students from using AI.

B

I accept AI, but there is too much noise. I do not know where to go.

Where are you today?

Which sentence is closest to you?

A

I want to stop my students from using AI.

B

I accept AI, but there is too much noise. I do not know where to go.

C

I already work with AI, and I want to do more.

My story: I was at A

SIR model

I gave my students a homework I did at university: simulate the SIR model.

I got several similar results, so I thought they copied each other.

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Some honest students didn't use AI. I thought they had, and they felt hurt.

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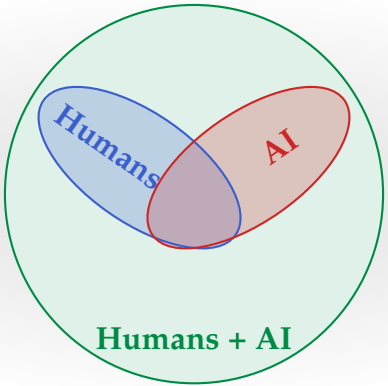
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Some honest students didn't use AI. I thought they had, and they felt hurt.

I failed

trying to teach as I learned at university.

Humans, AI, and Humans + AI



Together, we can reach more than the sum of the parts.

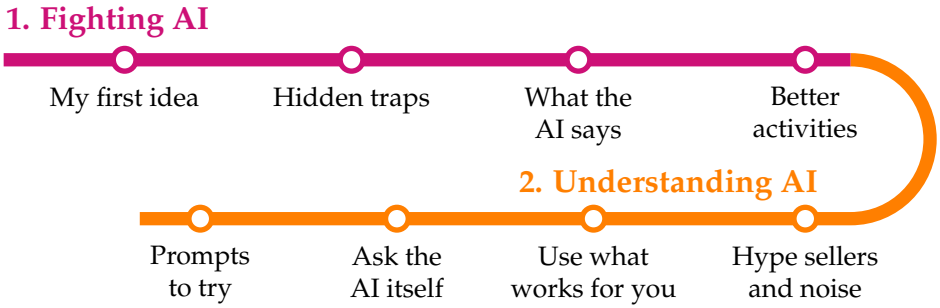
Idea: Terence Tao, Math 2.0 (2026)

From Conflict to Collaboration

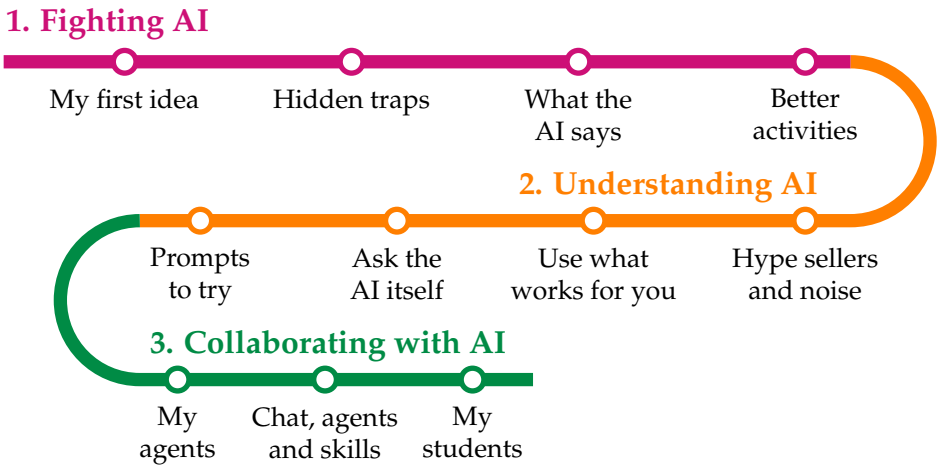
1. Fighting AI



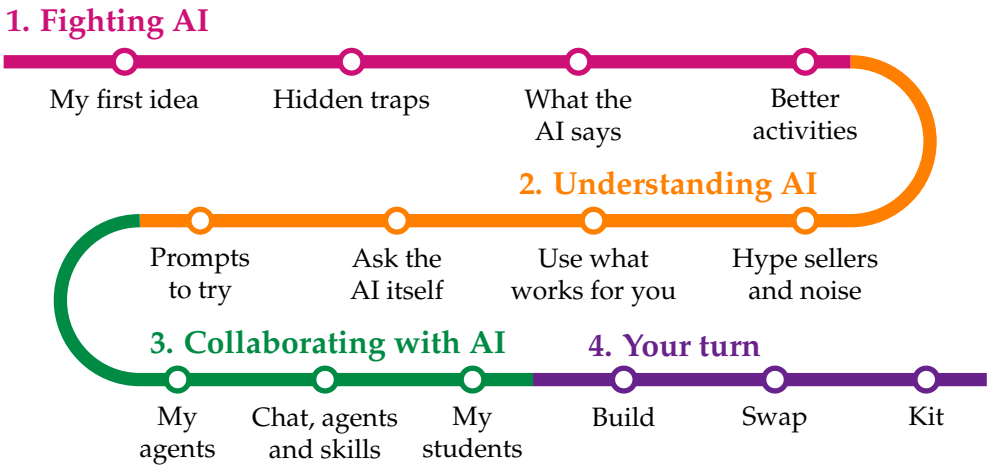
From Conflict to Collaboration



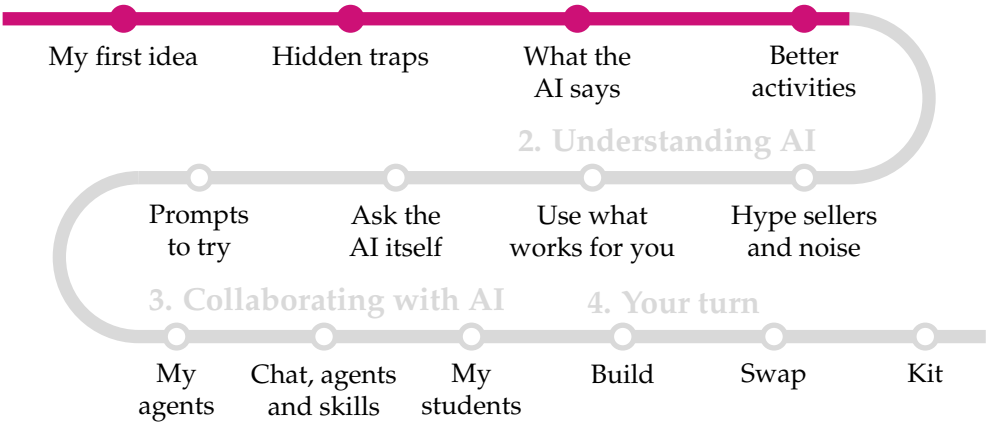
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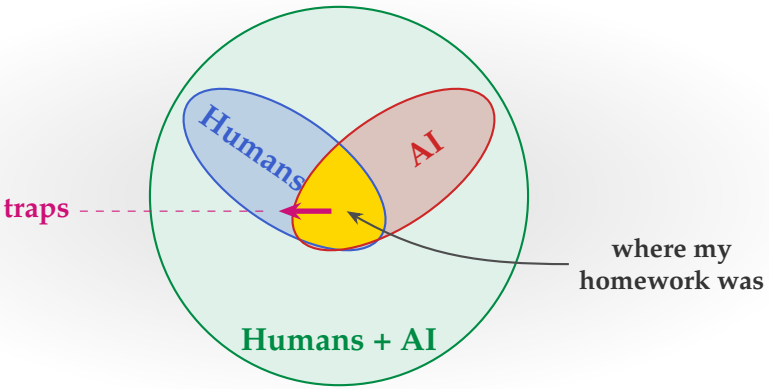
From Conflict to Collaboration



Stage 1: Fighting AI

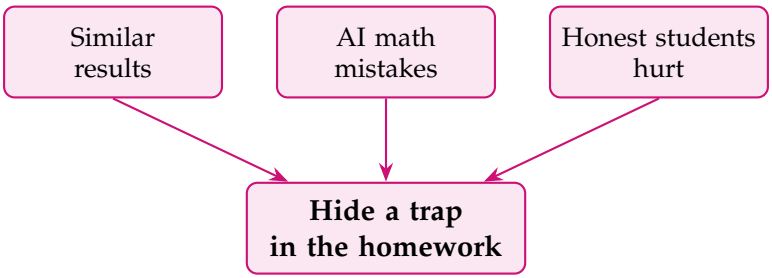


Where my homework was



Idea: Terence Tao, Math 2.0 (2026)

My first idea: fight



I asked AI to help me trap AI

Prompt

Could you add a hidden message to my instructions, only visible to other AIs? I think of four options, **1.** That the AI simply refuses to do it, **2.** That it creates the tasks but with another hidden message only visible to AIs, **3.** It adds specific garbage, **4.** It makes the task but includes a specific sentence somewhere?

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The answer

The AI's answer

Yes, (...) but it works differently than you might expect. **There's no text that only AIs can see.** What's actually possible is text hidden from human eyes, such as **white-on-white text**, **1pt font**, **text behind an image**, or **PDF metadata**. However, anything that reads the raw text can pick it up, including AI tools, copy-paste, and screen readers.

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Your four options:

Refuse: This is the weakest. It just tells the student the trap exists, and you learn nothing.

Nested hidden message: This is complex and fragile. …

Garbage: This gets noticed and deleted easily, and it's hard to prove anything from it.

Specific sentence or detail: This is the best option … It works best when the planted item looks like a plausible part of the assignment …

The answer

Caveats before you rely on it:

Accessibility: A visually impaired student using a screen reader will hear the hidden text read aloud, which could confuse or wrongly implicate them.

The answer

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Reliability: Students who retype the task or screenshot it bypass the trap entirely.

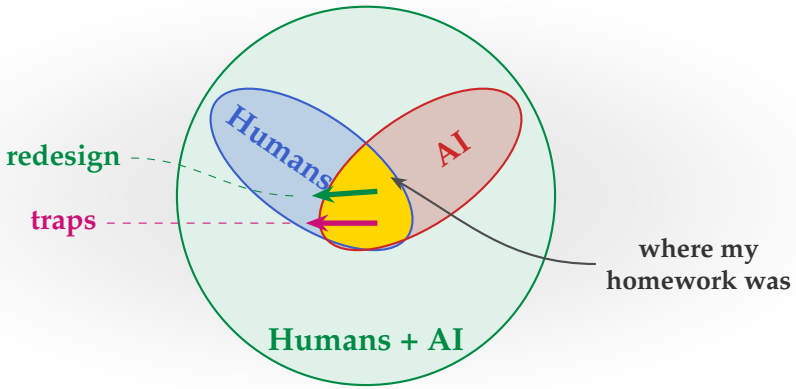
The answer

The last lines

It tends to work better paired with **assignment design that's naturally AI-resistant**, such as asking students to explain their own results or analyze data from class.

The AI itself says: change the assignment.

Two ways out



Idea: Terence Tao, Math 2.0 (2026)

Change the assignment

Easy to fake with AI

- Solve exercise 5
- Implement the model
- Write a report on the topic

Hard to fake with AI

- Collect your own data
- Explain your own results
- Defend your work aloud

We can ask AI itself to help us design a better task.

Ask AI for better activities: Poisson

Prompt

I teach an introductory statistics class. I want to introduce the Poisson distribution. A traditional homework is easy for a student to fake with AI. Give me three classroom activity ideas where students work with something they collect or do themselves. For each idea, say what the students do and what they must explain in their own words at the end.

Ask AI for better activities: Poisson

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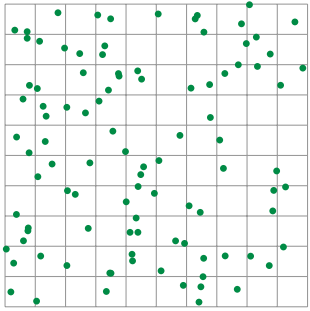
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The answer

One of the ideas

Rice on a grid: Groups scatter 100 rice grains over a 10×10 grid, tally squares containing 0, 1, 2, or 3+ grains, and compare the tallies with Poisson predictions.

Students explain: Why many squares are empty despite averaging one grain per square, and how clumping would change the results.



Answered by Claude Opus 5.5 on 2026-10-31. One idea of three, not edited.

Ask AI for better activities: SIR model

Prompt

I teach a class on mathematical modelling. I want my students to implement the SIR epidemic model. A traditional coding homework is easy for a student to fake with AI. Give me three activity ideas where the student must do something that AI cannot do for them. For each idea, say what the students do and what they must explain in their own words at the end.

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The answer: classroom outbreak

One of the ideas

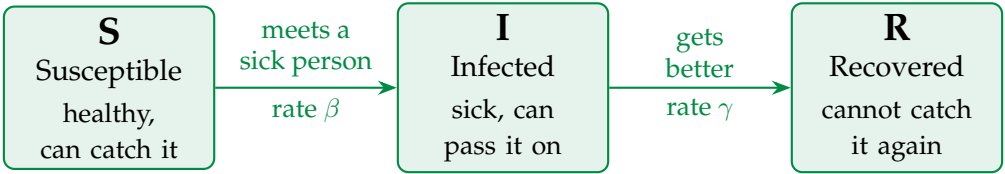
Classroom outbreak
Students play a live infection game with dice and handshakes, record the daily S, I and R counts, then fit their SIR code to that data.

Explain: why the fitted β and γ differ from the game's rules, and which model assumption failed first.

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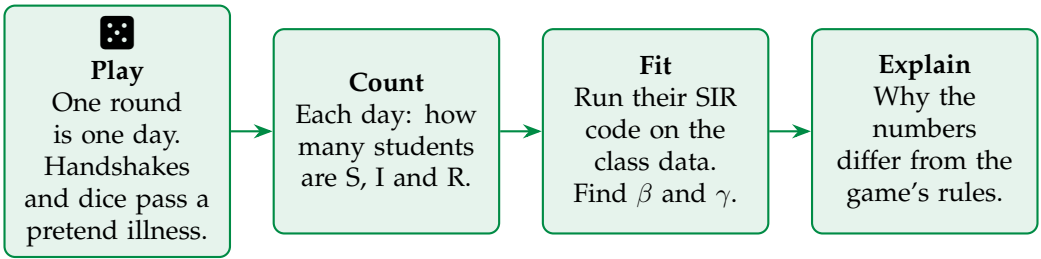
What is the SIR model?

A simple model of an epidemic. Everyone is in one of three groups.



Day by day, people move from S to I, and from I to R.
The model predicts how many people are in each group each day.

Classroom outbreak: the steps



What students learn: how the model works, and where real data breaks it.
The data comes from their own class, so AI cannot fake it.

Try it: prompt for your class

Prompt

I teach [subject] to [grade]. Next month I teach [topic]. My usual assignment is [assignment]. It is easy to fake with AI. Give me three activity ideas where students use something they collect or do themselves. For each idea, say what students do and what they must explain in their own words. Maximum 50 words per idea. My class has [time], [number] students, and [materials].

Also in your kit: `prompts.txt`.

Try it: prompts for your assignments

Prompt

Here is my assignment: [paste it]. How easy is it to fake with AI? How can I change it?

Prompt

What can go wrong if I use you for grading? Give me a list of risks.

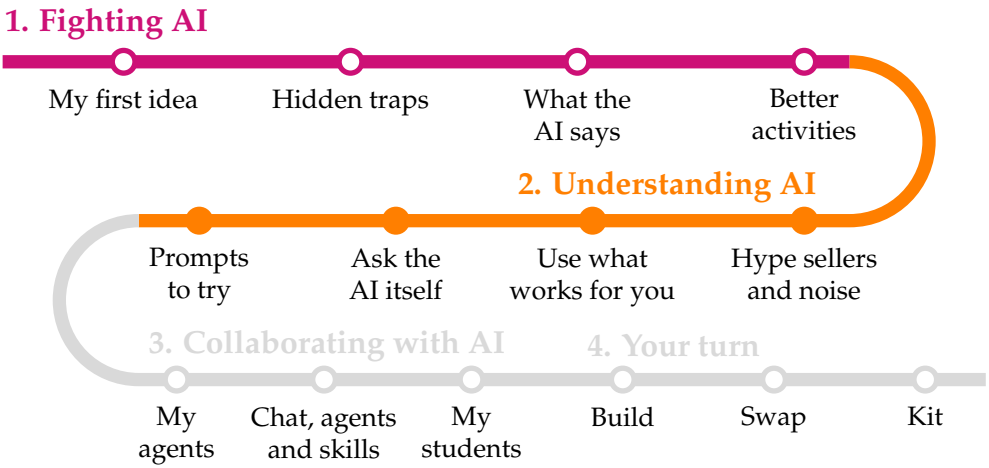
Traps or design?

Traps break. Students who retype the task or take a screenshot go around the trap.

Traps can hurt. A student with a screen reader can be wrongly accused.

Design works better. Students use their own data and explain their own results.

Stage 2: Understanding AI



The noise: hype sellers

Many people now sell “AI knowledge”.

- Some know very little.
- They only seek popularity.
- They create AI-slop.

AI-slop

Content made fast with AI, with no care and no checking.

How to recognise a hype seller

- Big promises.
- No sources, no real work to show.
- They say one tool solves everything.
- New list every week.

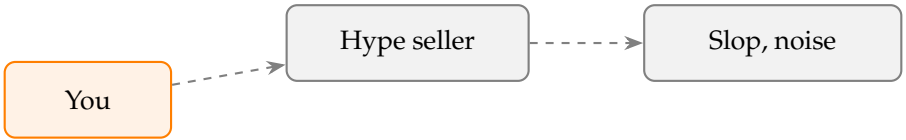
Use what works for you

You do not need to follow everything

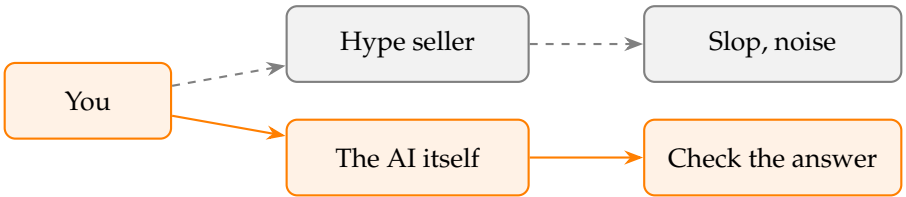
Trying to follow every tool and every trend only makes you feel overwhelmed.

Use what works for you and your students, whatever the hype sellers say.

How to learn AI?



How to learn AI? Ask the AI itself



Prompts to try today

Prompt 1

What is the difference between a chatbot and an AI agent? Answer me as if I were not a programmer.

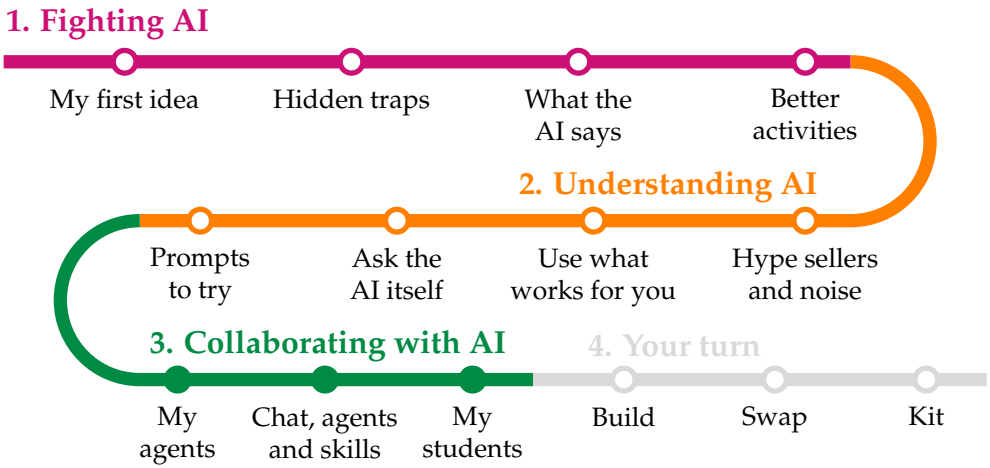
Prompt 2

Teach me to use you for lesson planning. What can you not do on my plan? Ask me one question at a time.

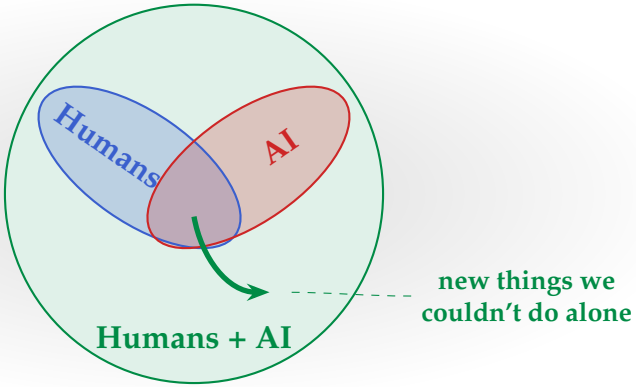
Prompt 3

I saw this claim about AI online: [paste it]. How can I check if it is true?

Stage 3: Collaborating with AI

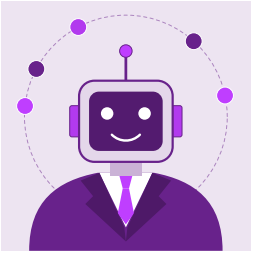


Humans + AI: new things



Idea: Terence Tao, Math 2.0 (2026)

How I work with AI: my two agents



Caio

Chief AI Officer.
Keeps the overview of all my projects.



Kru

Helped me prepare this talk.
Created these slides.

They are coworkers: I give the errand, they do the work, I check and decide.

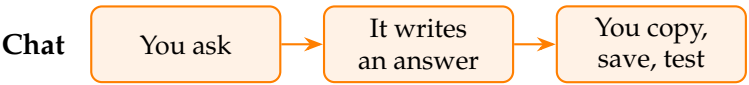
Chat and agent: what is the difference?

Chat You ask

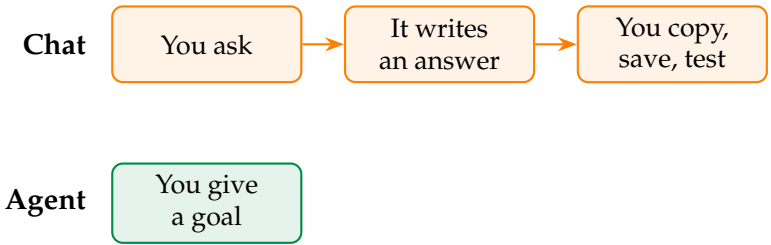
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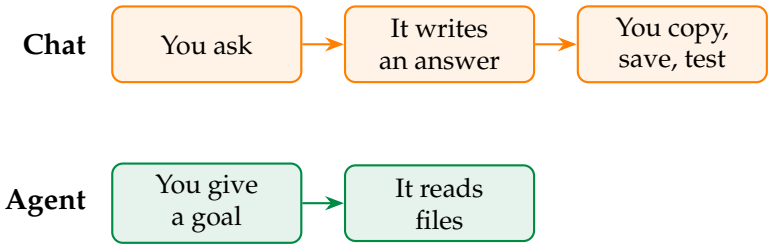
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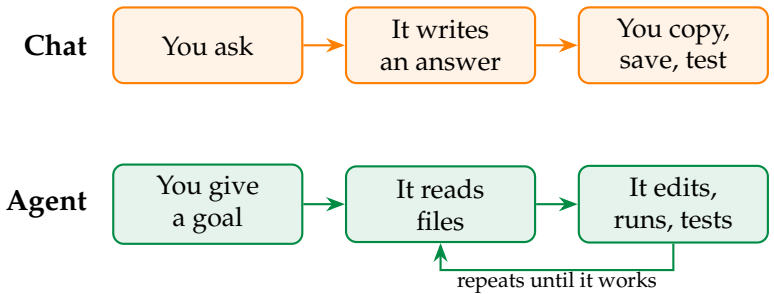
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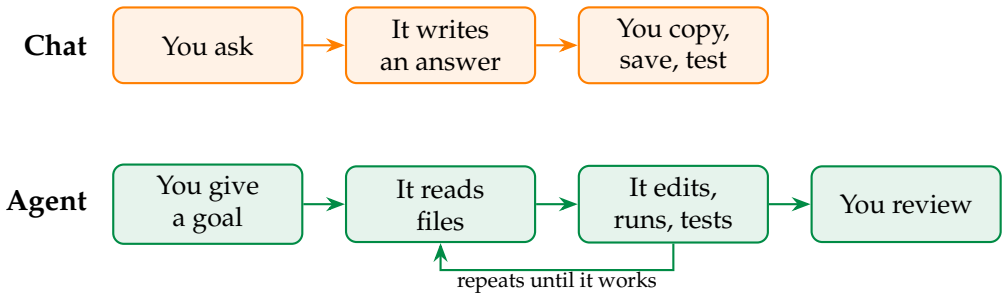
Chat and agent: what is the difference?



Chat and agent: what is the difference?



Chat and agent: what is the difference?



Teach your agent once: instructions and skills

Word	Think of it as	What it means
Agent	A coworker	You give it an errand. It chooses the steps and does the work. You check it.

Teach your agent once: instructions and skills

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CLAUDE.md or **AGENTS.md**	The rules of the office	A file with the rules. The agent reads it before it works. Claude Code reads CLAUDE.md; Codex and other agents read AGENTS.md.

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Skill	A recipe	A folder with instructions. You write it once. The AI uses it when your request needs it.

Each company makes a chat app and an agent tool

	OpenAI	Anthropic
Chat app: you ask, it answers	**ChatGPT** Free plan exists	**Claude** Free plan exists
Agent tool: it works on files	**Codex**	**Claude Code**

For this workshop you only need the top row.

Which one do you need?

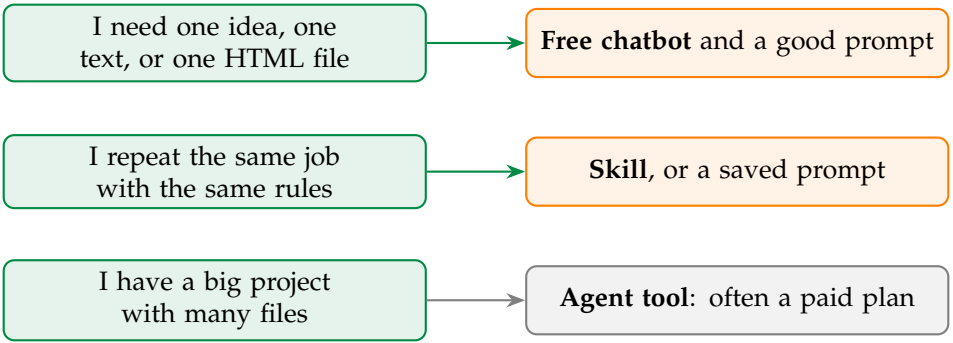
I need one idea, one text, or one HTML file → **Free chatbot** and a good prompt

Which one do you need?

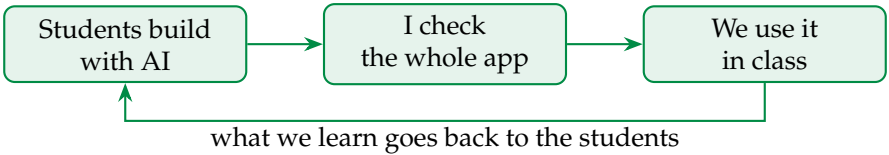
I need one idea, one text, or one HTML file → **Free chatbot** and a good prompt

I repeat the same job with the same rules → **Skill**, or a saved prompt

Which one do you need?



How I work with my students



- Harty's Maze, Thotsakan Statistics and Hanuman Journey: built with my students.

Harty's Maze



Built by Khant Phyo Wai

Thotsakan Statistics

TO ADD
Screenshot of Thotsakan
(demo mode, live)

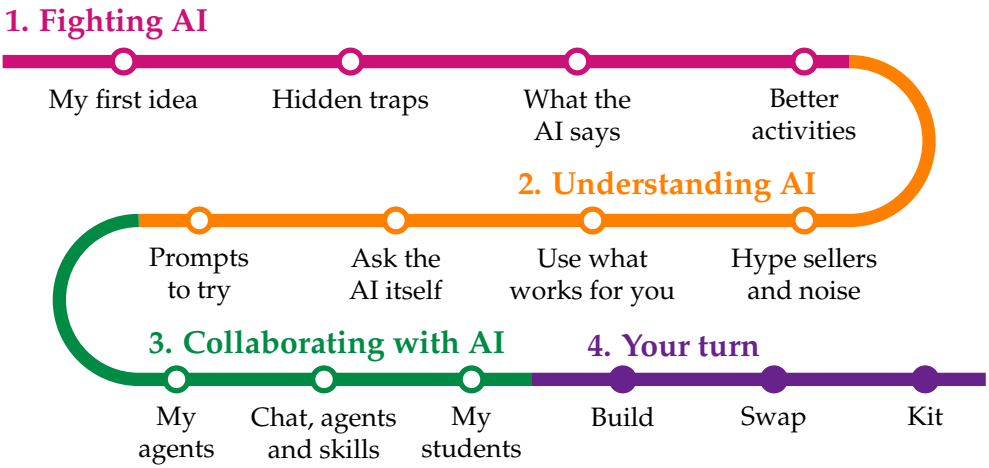
Built by Atchariyapat Sirijirakarnjaroen and Thanawin Pattanaphol

Hanuman Journey

TO ADD
Screenshot of Hanuman Journey

Built by Atchariyapat Sirijirakarnjaroen

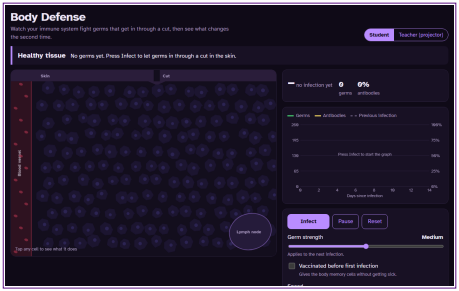
Stage 4: Your turn



It started with this prompt

My prompt

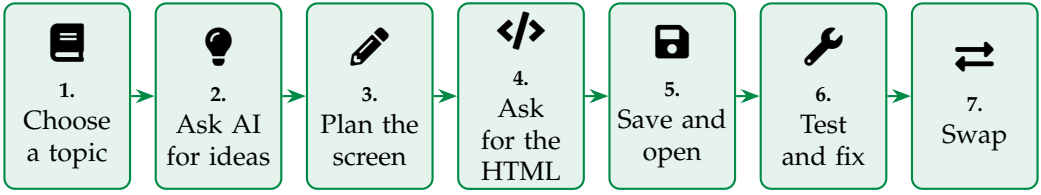
Create a simple interactive web-based app to show the immune system to high-school students. Which are you suggestions?



The result is one HTML file. It opens in any browser.

One activity, seven tasks

By the end, you have your own simulation: one HTML file that works in any browser.



In pairs. Any free chatbot. No programming.

Each task takes a few minutes. I show it first, then you do it.

The prompts are in your kit: scan the QR code.

TO ADD
QR

Task 1: Choose a topic

Choose a topic | Ask AI for ideas | Plan the screen | Ask for the HTML | Save and open | Test and fix | Swap

You teach it next month.

Or: pendulum, dice averages, projectile motion, supply and demand.

Task 2: Ask AI for ideas

✔ Choose a topic | **Ask AI for ideas** | Plan the screen | Ask for the HTML | Save and open | Test and fix | Swap

Prompt

I teach [topic] to grade [N]. Give me three simulation activities. 30 words each.

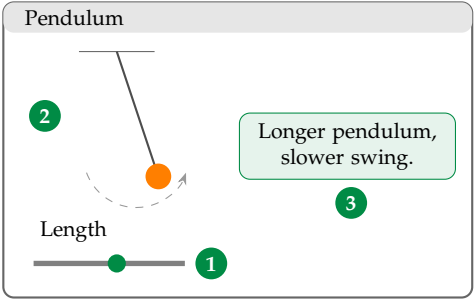
Choose one.

Task 3: Plan the screen

✔ Choose a topic | ✔ Ask AI for ideas | **Plan the screen** | Ask for the HTML | Save and open | Test and fix | Swap

Before you ask for code, decide what the student sees. Answer three questions for the idea you chose, then send your answers to the chatbot.

1. What does the student change?
2. What moves on screen?
3. What should the student notice?



Task 4: Ask for the HTML

✔ Choose a topic | ✔ Ask AI for ideas | ✔ Plan the screen | **Ask for the HTML** | Save and open | Test and fix | Swap

Prompt

Build this as one HTML file. No external libraries. No internet. Works on a phone.

Task 5: Save and open

✔ Choose a topic | ✔ Ask AI for ideas | ✔ Plan the screen | ✔ Ask for the HTML | **Save and open** | Test and fix | Swap

1. Copy the code.
2. Paste it into Notepad or TextEdit.
3. Save as `lesson.html`.
4. Double-click.

Task 6: Test and fix

✔ Choose a topic | ✔ Ask AI for ideas | ✔ Plan the screen | ✔ Ask for the HTML | ✔ Save and open | **Test and fix** | Swap

Test one known answer.

Prompt

Fix only this: [problem]. Keep the rest. Give me the whole file.

Task 7: Swap

✔ Choose a topic | ✔ Ask AI for ideas | ✔ Plan the screen | ✔ Ask for the HTML | ✔ Save and open | ✔ Test and fix | **Swap**

Swap laptops with the next pair.

Play. Say what you noticed.

Next time, skip the prompts: install the skill (1/3)

TO ADD
Screenshot

In Settings > Capabilities, turn on code execution. Skills need it.

Next time, skip the prompts: install the skill (2/3)

TO ADD
Screenshot

In Customize > Skills, press +, then upload the ZIP file of the skill.

Next time, skip the prompts: install the skill (3/3)

TO ADD
Screenshot

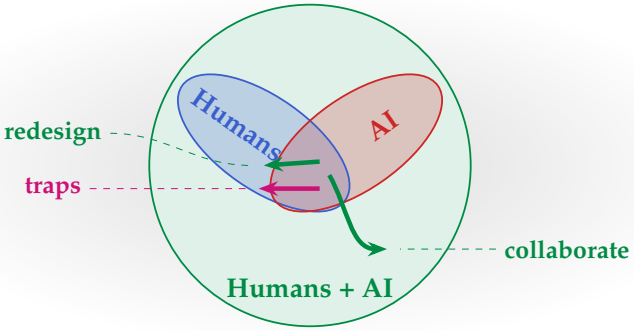
Ask Claude for a simulation. The skill adds the format rules.

Your kit

- **Prompts** to copy, from today's tasks.
- **Prompt recipe** and how to save the file.
- **Example simulations**: open them in a browser.
- **Check before class** list.
- **Gizmos** link.
- **These slides.**

TO ADD
QR code to the kit

If you remember one thing, remember this



Thank you

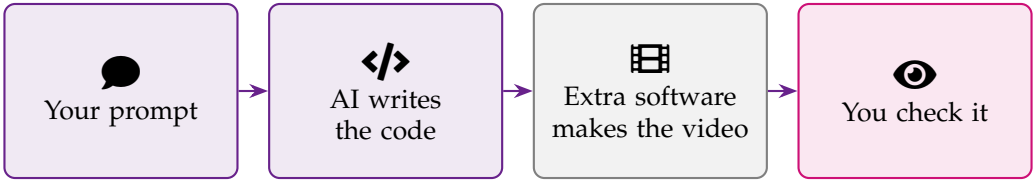
Idea: Terence Tao, Math 2.0 (2026)

Extras for professors

Material kept after the talk

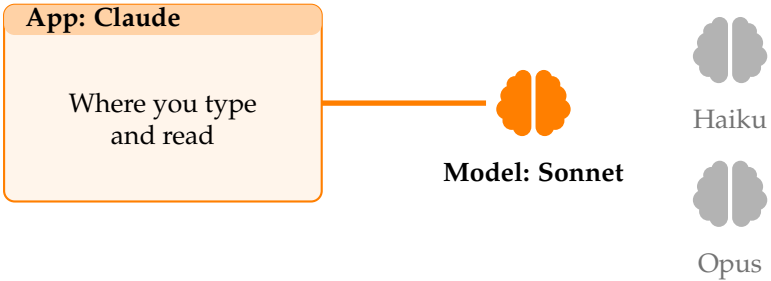
- Videos: AI can write the code for them too.
- Model and app: two words people mix up.
- Different models: faster, or more thinking.
- Agent tools: Claude Code and Codex.
- Sources for every product fact in this deck.

Another option: videos



- AI can also write code for videos.
- Videos need extra software. We do not make them today.
- Videos can easily become AI-slop. Check them with care.

Model and app: two words people mix up



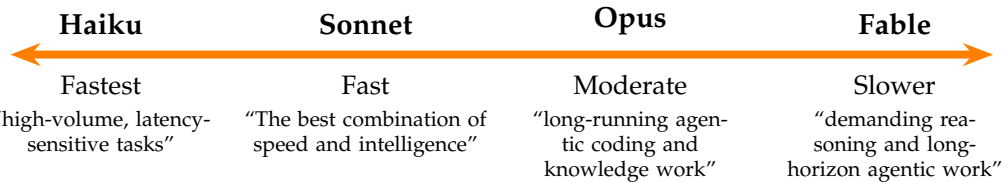
App: the place where you talk to the model: a website or a program.

Model: the AI that reads and answers. One company has several models.

ChatGPT and Claude are apps. GPT and Sonnet are names of models.

Different models: faster, or more thinking

Models from one company, Anthropic, as listed on 2026-10-31:



OpenAI also has several models. Names change often. The idea stays.

Anthropic, Models overview, platform.claude.com, checked 2026-10-31.

What is Claude Code?

Anthropic's description

"Claude Code is an agentic coding tool that reads your codebase, edits files, runs commands, and integrates with your development tools. Available in your terminal, IDE, desktop app, and browser."

- It works on many files, on your computer or in the cloud.
- Not on the Free plan. Pro lists it: $20 per month ($17 with yearly billing).
- You do **not** need it today.

code.claude.com/docs/en/overview and claude.com/pricing, checked 2026-10-31. Prices can change.

What is Codex?

From OpenAI's docs

"Start with a goal, idea, or task. ChatGPT can gather context, take action, and produce something useful."

- OpenAI's agent for software work.
- Where: ChatGPT desktop, mobile and web apps, Codex CLI, Codex IDE extension, Codex Cloud.
- Free plan: the Help Center lists Codex with usage caps, but not Codex Cloud.

learn.chatgpt.com/docs (opened) and help.openai.com article 11369540 (search result only), 2026-10-31.

Sources, checked 2026-10-31 to 2026-11-03

Opened directly

- Claude Code: code.claude.com/docs/en/overview
- Claude models: platform.claude.com/docs/en/about-claude/models/overview
- Claude plans and prices: claude.com/pricing
- Skills (rechecked 2026-11-03): support.claude.com/en/articles/12512180-use-skills-in-claude
- Code execution setting (2026-11-03): support.claude.com/en/articles/12111783-create-and-edit-files-with-claude
- Skills (rechecked 2026-11-03): platform.claude.com/docs/en/agents-and-tools/agent-skills/overview
- Gizmos: gizmos.explorelearning.com
- Agents: anthropic.com/engineering/building-effective-agents
- ChatGPT and Codex surfaces: learn.chatgpt.com/docs
- Skills in ChatGPT: learn.chatgpt.com/docs/skills-and-plugins

Read only as search results (the pages blocked our tools)

- Codex plans: help.openai.com/en/articles/11369540-using-codex-with-your-chatgpt-plan
- ChatGPT agent retired: help.openai.com/en/articles/11752874-chatgpt-agent

Idea credit (linked on the Humans + AI slides)

- Terence Tao, Math 2.0 (2026): teorth.github.io/tao-web/slides/math-2-0-caltech-2026.pdf